

Multi-Subnet Network Configuration & Web/DNS Implementation

Implementation, Routing, and Domain Name Resolution using Cisco Packet Tracer

Author Name:	Iqra Jamil
University:	University of Management and Technology (UMT)
Domain / Subject:	Computer Networking & Inter-VLAN / Router Configuration
Tool Used:	Cisco Packet Tracer (v9.0.1)
Target Audience:	Network Engineers & Technical recruiters on LinkedIn

1. Executive Summary & Overview

This lab technical documentation illustrates the end-to-end configuration, subnetting, and service integration of a dual-subnet enterprise-style network modeled inside Cisco Packet Tracer. The lab scenario demonstrates how network endpoints across different broadcast domains route requests through a central router (2811), perform DNS lookup queries, and successfully communicate over HTTP/HTTPS with configured server nodes.

Key Architectural Highlights:

- **Subnet 1 (192.168.1.0/24):** Hosts end-user clients (`PC0`, `PC1`) along with direct Web Servers (`gmail.com` and `google.com`).
- **Subnet 2 (192.168.2.0/24):** Hosts endpoint devices (`PC2`, `PC3`) alongside the dedicated Centralized DNS Server.
- **Inter-Network Routing:** Managed via Cisco 2811 Router linking Switch 0 and Switch 1.
- **Services Implemented:** Domain Name System (DNS A-Records) mapping domain names to IP endpoints, and Web Server HTTP stack configuration.

This project proves hands-on proficiency in IP addressing scheme design, gateway configuration, server side service provisioning, and client-side browser testing.

2. Overall Network Topology Architecture

The complete logical topology consists of two distinct Local Area Networks (LANs) connected via a Central Dual-FastEthernet Router (Cisco 2811). The left segment operates on the 192.168.1.0/24 subnet while the right segment operates on the 192.168.2.0/24 subnet.

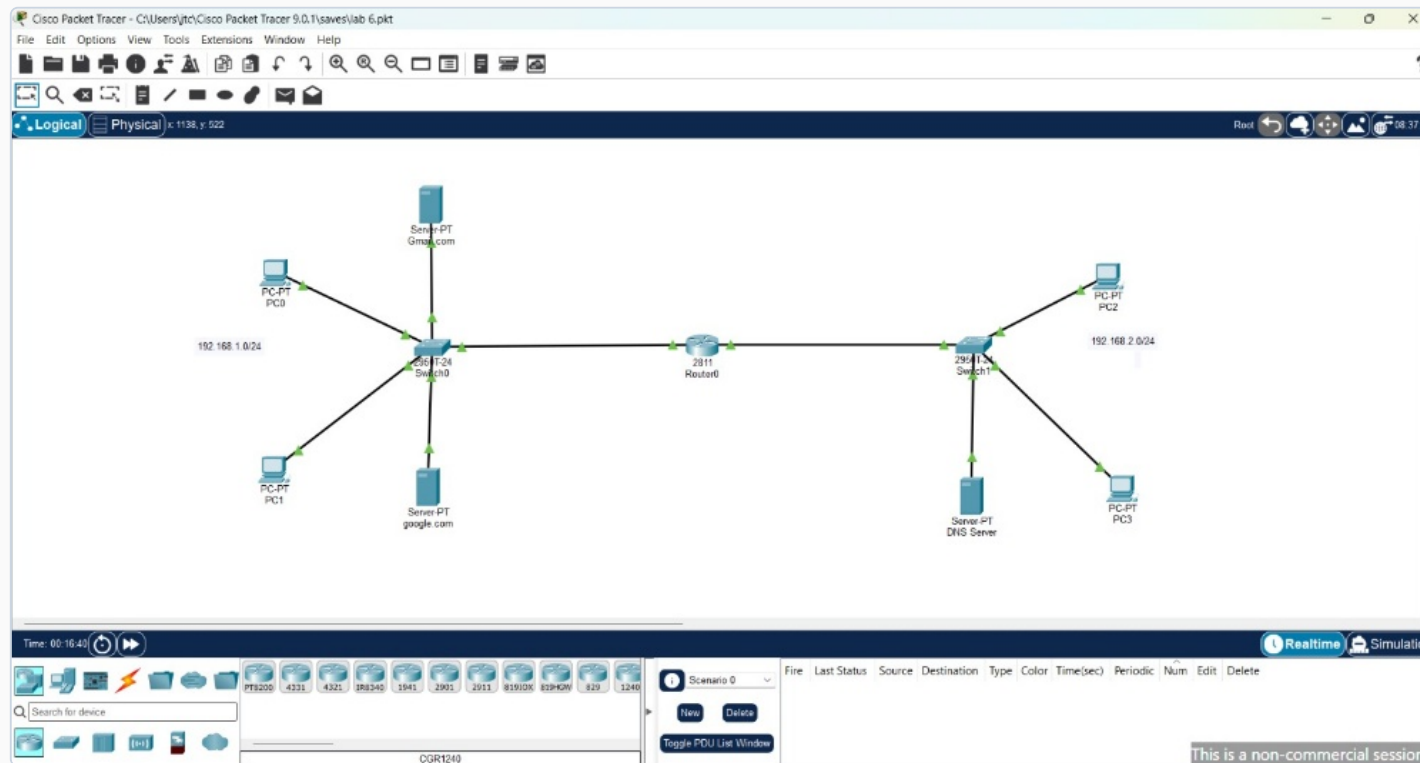


Figure 1: Complete Cisco Packet Tracer Network Topology layout showing Subnets 192.168.1.0/24 and 192.168.2.0/24 linked by Cisco 2811 Router.

3. Centralized DNS Server Network Settings

The Central DNS Server is situated on the secondary subnet with static IP address 192.168.2.10, subnet mask 255.255.255.0, and default gateway set to 192.168.2.3 (Router interface). Correct gateway assignment is mandatory so client nodes across subnets can reach DNS services.

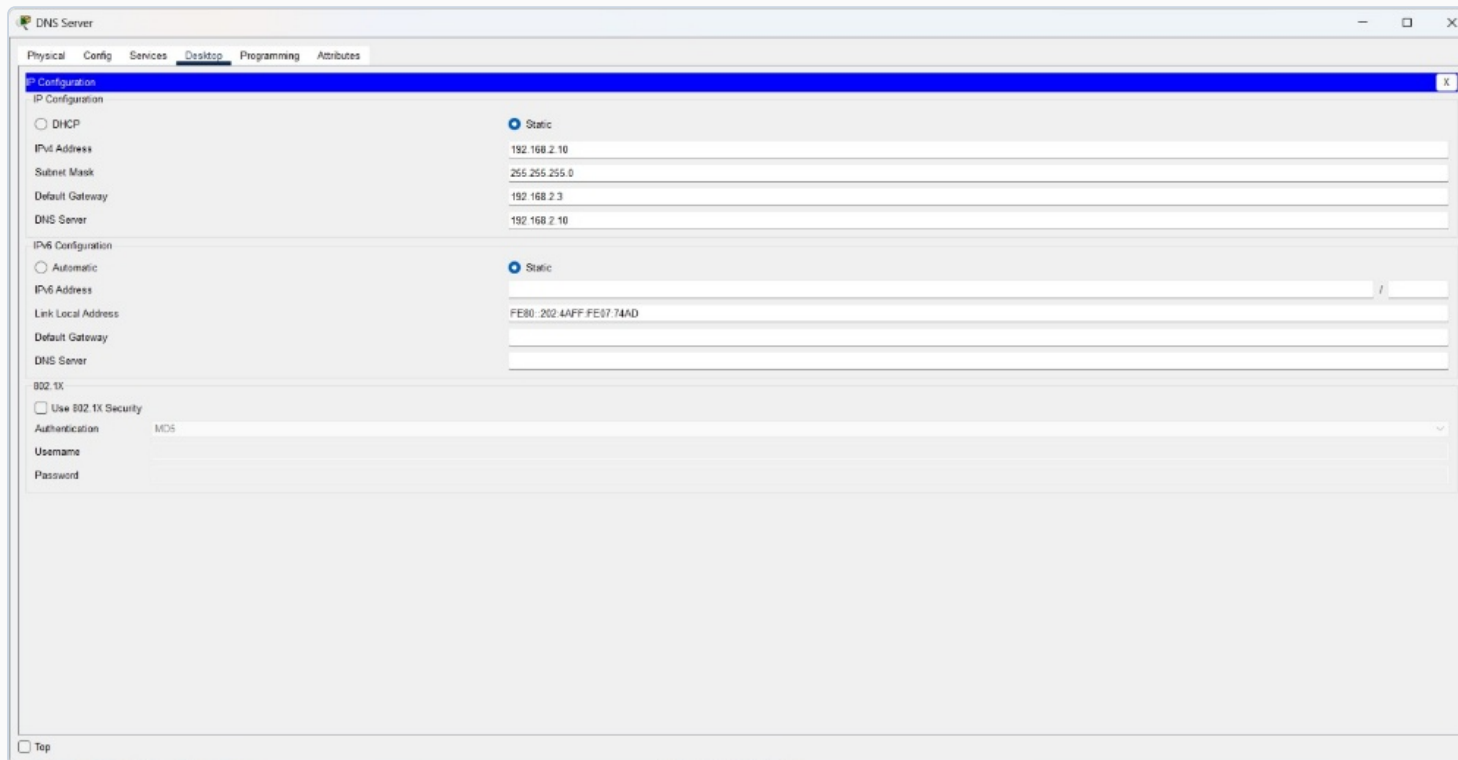


Figure 2: Static IP, Subnet Mask, Gateway, and Self-DNS Address settings for the DNS Server node.

4. DNS Records & Hostname Resolution Configuration

The DNS Service on the DNS Server is activated and configured with Resource A-Records. These entries map human-readable URLs to their corresponding web server IPv4 destinations:

- gmail.com → 192.168.1.5
- google.com → 192.168.1.10

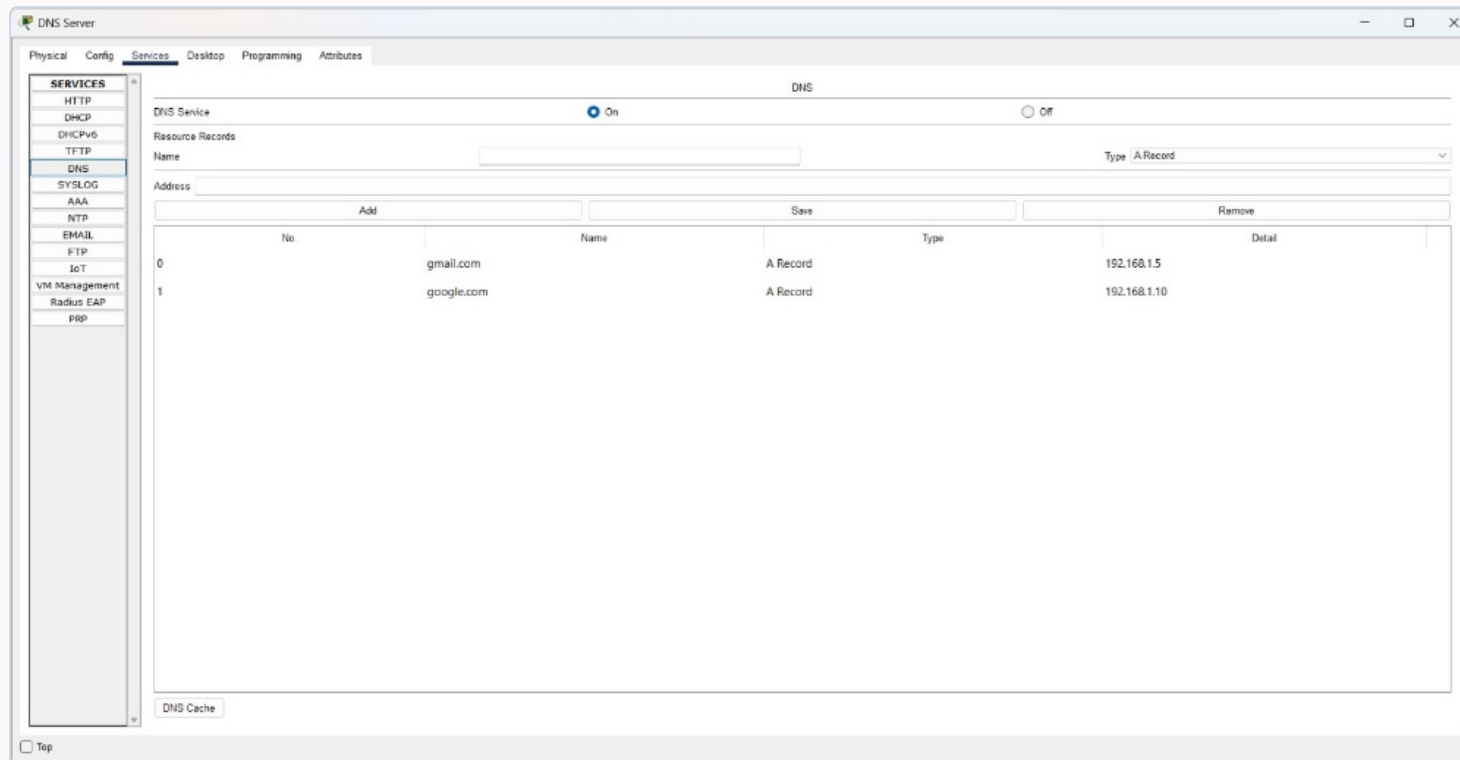


Figure 3: Active DNS Records list configured on the DNS Server showing domain mapping to web servers.

5. Web Server Configuration (Gmail.com Server)

The Web Server hosting gmail.com is statically assigned IP address 192.168.1.5 with Default Gateway set to 192.168.1.1 and Primary DNS pointing to 192.168.2.10 across the router link.

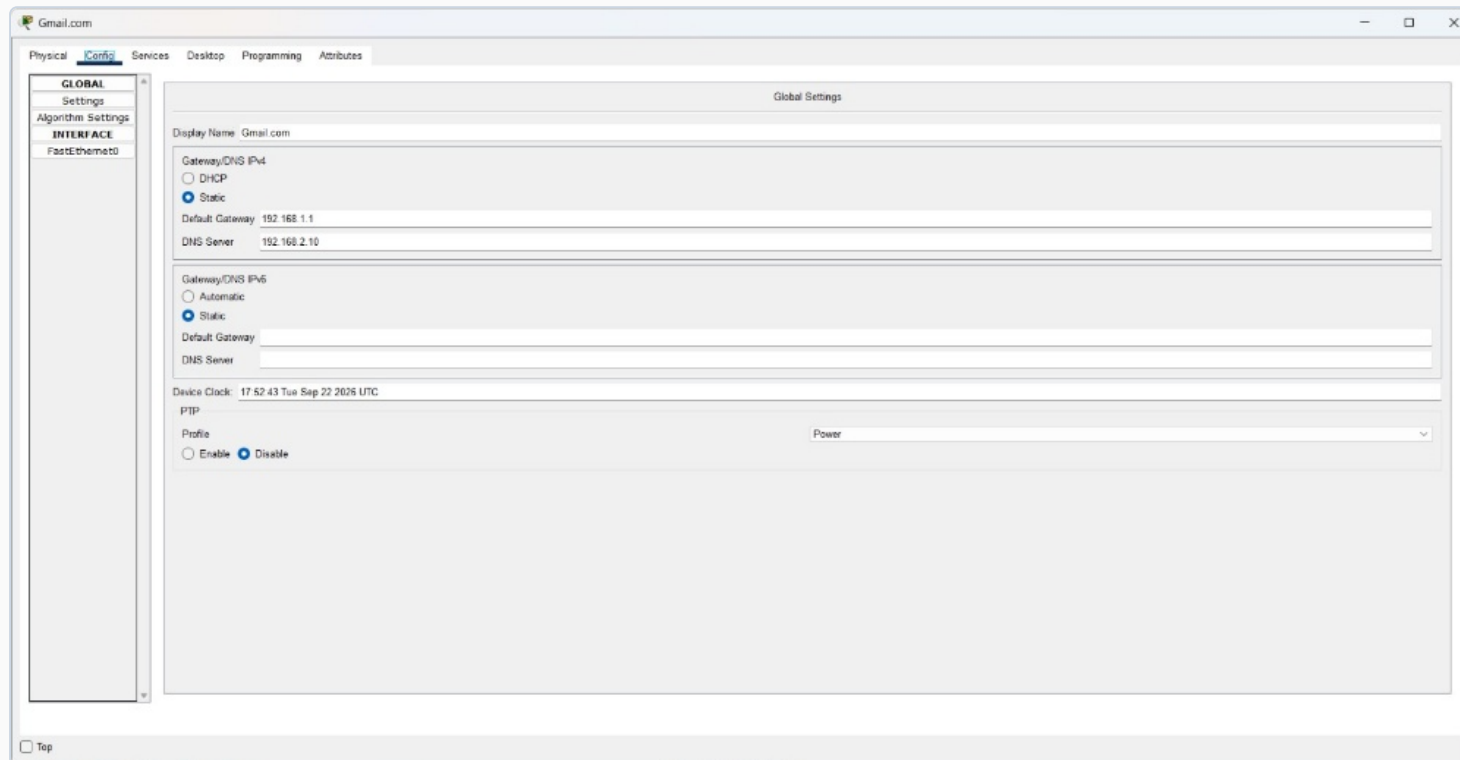


Figure 4: Global configuration menu for the Gmail server establishing default gateway and remote DNS server pointers.

6. DNS Service Verification & Sync

Ensuring domain entries are properly saved and actively listening on port 53. The DNS table entries guarantee cross-subnet reachability whenever client requests arrive from PC endpoints.

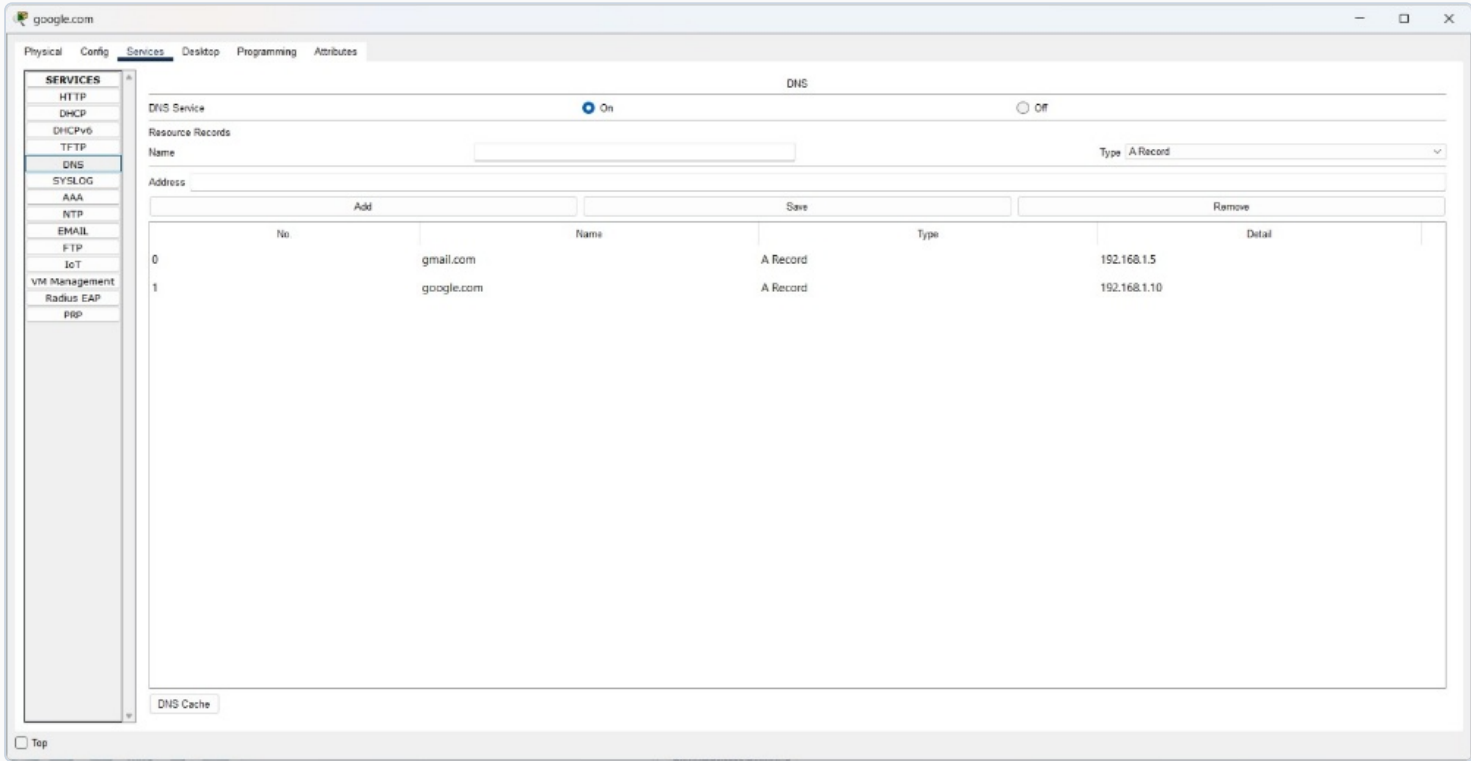


Figure 5: Detailed view of the active DNS Service configuration panel verifying entry state.

7. Functional Testing: Client Browser Accessing Gmail.com

To verify successful setup, endpoint PC0 initiates an HTTP GET request by entering `http://gmail.com` in its built-in Web Browser. The client queries `192.168.2.10`, resolves `192.168.1.5`, and renders the hosted web page seamlessly.

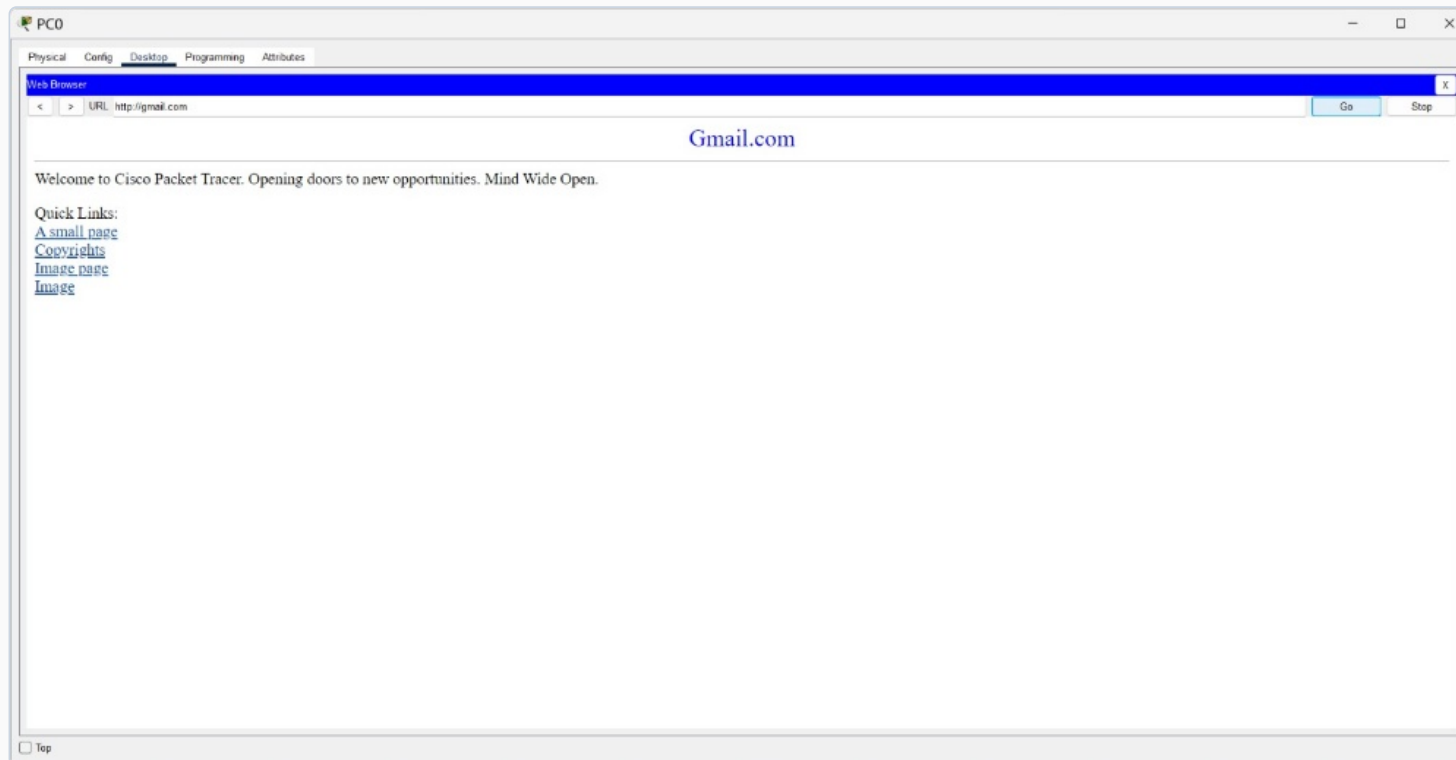


Figure 6: Client PC0 successfully loading `http://gmail.com` via DNS hostname resolution.

8. Functional Testing: Client Browser Accessing Google.com

Subsequent client request testing for `http://google.com` performed from PC0. The DNS server resolves the query to IP address 192.168.1.10, demonstrating robust multi-server resolution across subnets.

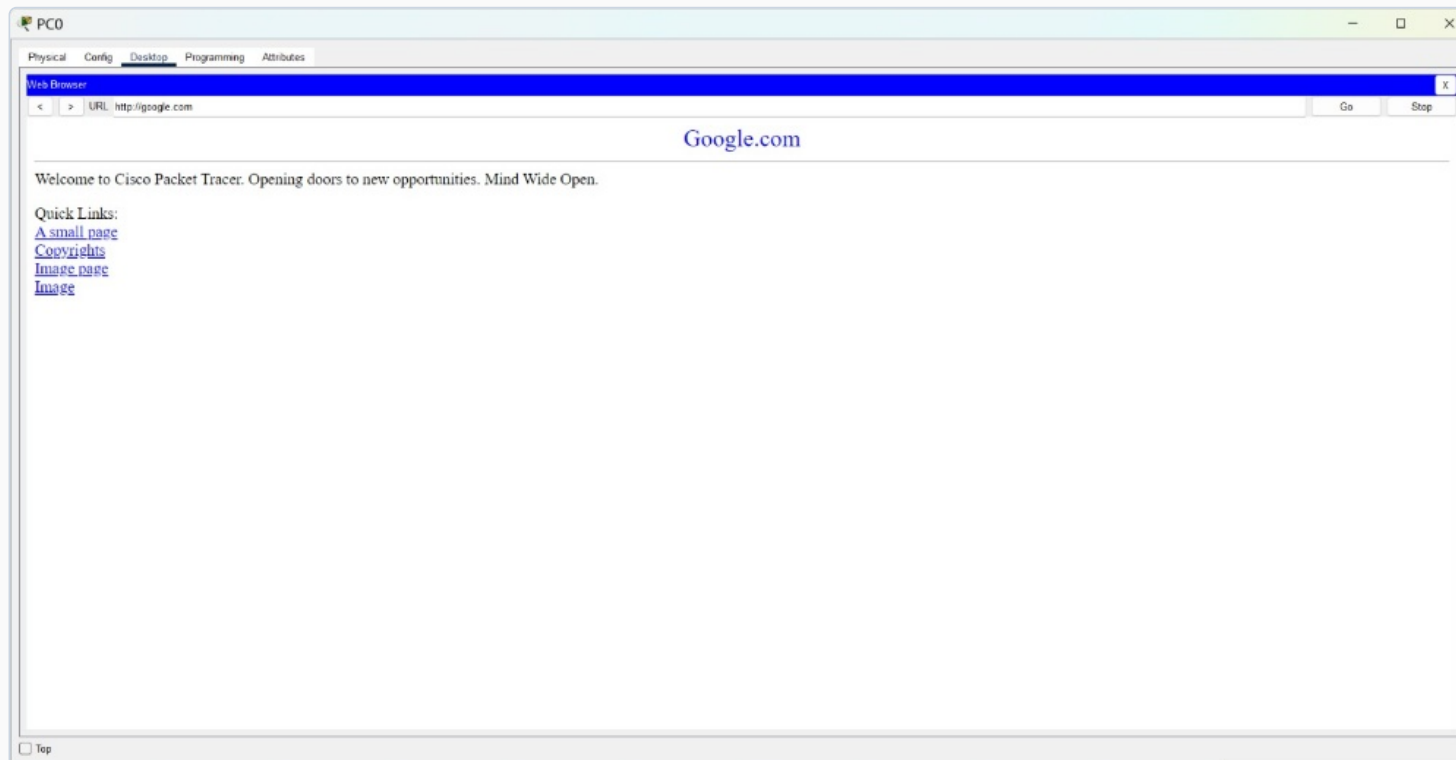


Figure 7: Client PC0 successfully retrieving `http://google.com` HTML content via Cisco Packet Tracer browser.

9. Conclusion & Key Technical Takeaways

This simulation lab confirms practical understanding of foundational computer networking concepts essential for network engineering and security roles:

Skills Demonstrated:

- **IP Subnetting & Addressing:** Designed and assigned static IPv4 parameters across multiple subnet domains (`192.168.1.0/24` and `192.168.2.0/24`).
- **Cross-Subnet Gateway Routing:** Interconnected distinct switch domains using a Cisco 2811 series router with proper gateway endpoints.
- **DNS Name Resolution:** Deployed a centralized DNS server, created A-records, and verified name resolution for multiple domains.
- **Application Layer Services (HTTP/DNS):** Configured HTTP web servers and verified client-server connectivity via embedded browser utilities.

Prepared By: Iqra Jamil

Institution: University of Management and Technology (UMT)

Document Purpose: Professional Portfolio & LinkedIn Technical Showcasing